

Name: _____

Class: _____

Date: _____

Homework 5.1

Beneficial and Harmful Effects

Score: ____ / 8

Multiple Choice

1. An **unintended** effect of a computing innovation is...

- A. always harmful
- B. an effect its creators did not plan for
- C. always beneficial
- D. a syntax error

2. Which statement is true?

- A. Every effect of an innovation is either good for everyone or bad for everyone
- B. Harmful effects only come from bad code
- C. Innovations only have intended effects
- D. The same effect can be beneficial to some people and harmful to others

3. Social media connects friends but can also spread misinformation quickly. This shows that...

- A. only harmful effects matter
- B. social media has no effects
- C. misinformation is intended
- D. an innovation can have both beneficial and harmful effects

4. Responsible programmers...

- A. ignore possible harms
- B. consider how their innovations could be used in unintended ways
- C. only care about speed
- D. never update their programs

Written 1

Choose a computing innovation. Describe one beneficial effect, one harmful effect, and identify who is affected by each.

Written 2

Describe one way a developer could reduce a harmful effect of a social media app.

Name: _____

Class: _____

Date: _____

Homework 5.2

Digital Divide

Score: ____ / 8

Multiple Choice

1. The digital divide is...
 - A. the gap between two computers
 - B. a type of network cable
 - C. differing access to computing devices and the Internet based on socioeconomic, geographic, or demographic characteristics
 - D. a compression method
2. A rural town with no broadband service is an example of which digital divide factor?
 - A. Demographic
 - B. Socioeconomic
 - C. Geographic
 - D. Technical
3. A family that cannot afford a laptop illustrates which factor?
 - A. Geographic
 - B. Legal
 - C. Demographic
 - D. Socioeconomic
4. Older adults being less comfortable with new technology relates to which factor?
 - A. Socioeconomic
 - B. Geographic
 - C. Demographic
 - D. Political

Written 1

A school assigns homework that requires streaming videos. Explain how this could affect students differently and suggest two solutions.

Written 2

Explain why owning a smartphone doesn't fully close the digital divide.

Name: _____

Class: _____

Date: _____

Homework 5.3

Computing Bias

Score: ____ / 8

Multiple Choice

1. Bias in computing innovations can come from...
- A. using too many variables
 - B. only hardware failures
 - C. biases in the data used or written into the algorithms
 - D. compression
3. A facial recognition system trained mostly on images of lighter-skinned people is likely to...
- A. be less accurate for people with darker skin
 - B. be equally accurate for everyone
 - C. work only at night
 - D. never make mistakes

2. What is the best way to reduce bias in a machine learning model?
- A. Remove all testing
 - B. Use a faster computer
 - C. Collect more data from the same source
 - D. Train it on data that represents the full population it will serve
4. Does collecting MORE data from the same biased source remove bias?
- A. Yes, always
 - B. Only if the data is compressed
 - C. No, the bias remains
 - D. Only for small data sets

Written 1

A bank uses an algorithm trained on 30 years of past loan decisions. Explain how bias could enter and give two ways to reduce it.

Written 2

Explain how a diverse development team could help reduce bias. Give a specific example.

Name: _____

Class: _____

Date: _____

Homework 5.4

Crowdsourcing

Score: ____ / 8

Multiple Choice

1. Crowdsourcing is...

- A. a sorting algorithm
- B. a type of encryption
- C. a cloud storage device
- D. obtaining input or information from a large number of people via the Internet

2. Which is an example of citizen science?

- A. Volunteers recording bird sightings in an app used by researchers
- B. A scientist working alone in a lab
- C. A company selling telescopes
- D. Watching a science video

3. Raising money for a project from many people online is called...

- A. crowdfunding
- B. phishing
- C. routing
- D. sampling

4. What is a main benefit of crowdsourcing?

- A. Only experts participate
- B. Guaranteed accuracy
- C. No Internet needed
- D. Collecting large amounts of data quickly

Written 1

Design a crowdsourcing project for your school. Describe what people contribute, one benefit, and one challenge.

Written 2

Explain how the digital divide could affect the results of a crowdsourced project.

Name: _____

Class: _____

Date: _____

Homework 5.5

Legal and Ethical Concerns

Score: _____ / 8

Multiple Choice

1. Intellectual property (IP) is...

- A. computer hardware
- B. any data on the Internet
- C. material created by a person, such as code, music, or writing, that belongs to its creator
- D. public domain only

2. A Creative Commons license allows...

- A. anyone to claim the work as their own
- B. creators to let others share, use, and build on their work under certain conditions
- C. only the government to use a work
- D. no sharing at all

3. Open-source software...

- A. can never be changed
- B. can be freely shared and modified
- C. is always expensive
- D. has no code

4. Using someone else's work without credit and presenting it as your own is...

- A. plagiarism
- B. encryption
- C. compression
- D. crowdsourcing

Written 1

A student wants to use a song, a photo, and some code from the Internet in a game they will publish. For each, describe how they could use it legally and ethically.

Written 2

Explain the difference between legal and ethical, using a computing example.

Name: _____

Class: _____

Date: _____

Homework 5.6

Safe Computing

Score: _____ / 8

Multiple Choice

1. Which is personally identifiable information (PII)?

- A. The weather
- B. A Social Security number
- C. A public library's address
- D. A random number

2. Phishing is...

- A. a technique that tricks users into providing personal information
- B. a type of compression
- C. a backup method
- D. a routing protocol

3. Multifactor authentication requires...

- A. no password
- B. two passwords
- C. at least two different types of evidence, such as a password and a code from a phone
- D. a longer username

4. Encryption is...

- A. sorting data
- B. deleting data
- C. compressing data
- D. encoding data to prevent unauthorized access

Written 1

An email says: "Your account will be closed! Click here and enter your password within 24 hours." Identify the attack, two warning signs, and what the user should do.

Written 2

Explain how public key encryption lets two people communicate securely without ever sharing a secret key.
